Team 14 Design Document: LitClub

<u>Team Members:</u>
Kathleen Lowe, Ben Herrington, Sofia Hughes, Brayden Reimann, Vidya Madana

Index

Purpose

While there are sites online for users to discuss their favorite pieces of literary media, few replicate the experience of going to a book club in person. In a book club, people can discuss novels as they go along; no spoilers are being discussed because the understanding is that everyone has read up to a certain point. With LitClub, users can connect with a reading community at their own pace. Our app serves as a platform for passionate readers to connect with others who are reading the same books as them, interested in similar genres, and have their own recommendations to share. Along with a chapter-by-chapter reading forum, our app will also have a dedicated social network for users to share updates about what they're reading, share book recommendations, and more. Through our app, readers around the world can connect with one another and discuss their current reads.

Functional Requirements:

1. As a reader, I would like to create a user profile with a username, password, profile image, and a nickname.
2. As a reader, I would like to add a short biography to my profile.
3. As a reader, I would like to select what genres of books I'm interested in seeing recommended to me.
4. As a reader, I would like to select books I am currently reading to put on my shelf.
5. As a reader, I would like to return to books on my shelf to go to their forums.
6. As a reader, I would like to bookmark pages/chapters to return to.
7. As a reader, I would like to checkmark chapters to access the forum for that chapter.
8. As a reader, I would like to modify the status of a given book between four statuses: Want to read, Currently reading, On hiatus, or Finished.
9. As a reader, I would like to be able to choose what reading information I share publicly, such as current reads. I should have the option to hide a book from my public bookshelf.
10. As a reader, I would like to be able to follow other readers.
11. As a reader, I would like to view my followers and following users.
12. As a reader, I would like to direct messages to other users.
13. As a reader, I would like to be able to block other users.
14. As a reader, I would like to save or "dogear" books I want to eventually read.
15. As a reader, I would like to create personalized bookshelves depending on genre or a category of my choosing to compile books into specific reading lists.
16. As a reader, I would like to be able to post blog entries onto my main page without those posts being a part of a given forum.
17. As a reader, I would like other readers to be able to view my past book reviews or forum comments from my profile page.

18. As a reader, I would like to view forums that correspond to each chapter of the book I am reading.
19. As a reader, I would like to share my thoughts by making a comment on a forum.
20. As a reader, I would like to make subcomments under any other comment or subcomment.
21. As a reader, I would like to have the ability to upvote/like comments I agree with.
22. As a reader, I would like to be notified when I receive subcomments from other users.
23. As a reader, I would like to be able to report comments or posts that I find offensive.
24. As a reader, I would like for comments flagged as having spoilers for chapters beyond the scope of the specific forum to be censored or blurred.
25. As a reader, I would like to mark if my chapter review is from a first-time read or a reread.
26. As a reader, I would like to review the book as a whole.
27. As a reader, I would like to mark my review as having or not having spoilers before publishing my review.
28. As a reader, I would like to rate my books out of 5 stars based on several fixed categories to calculate my overall averaged review.
29. As a reader, I would like to flag reviews posted by other users that have spoilers.
30. As a reader, I would like to add tags for my reviews that describe my feelings on the book.
31. As a reader, I would like to sort and read reviews by tags.
32. As a reader, I would like to see the average rating of a book based on others' reviews.
33. As a reader, I would like to sort and read reviews by rating.
34. As a reader, I would like to choose which of my reviews, if any, will be displayed/spotlighted on my user profile.
35. As a reader, I would like to be able to navigate to someone's profile from the review they posted.
36. As a reader, I would like to be able to submit summaries for a chapter, with the potential for that summary to be displayed at the top of the chapter's forum.
37. As a reader, I would like to upvote or downvote potential summaries to decide which one gets the forum spotlight.
38. As a reader, I would like to be able to join a book club so that I can read books along with others.
39. As a book club leader, I would like to be able to create a book club that readers can join.
40. As a book club leader, I would like to create a list of books for my book club to read.
41. As a book club leader, I would like to select a book for my club to be currently reading.
42. As a book club leader, I would like to create discussion threads specific to my book club.
43. As a book club leader, I would like members to be able to suggest and vote on the next book for the club to read.

44. As a book club leader, I would like to make my book club publicly discoverable and joinable by anyone, or choose to make it private and available only via link.
45. As a book club member, I would like to access discussion pages that my book clubs have set up.
46. As a book club member, I would like to view past books that book clubs have read.
47. As a book club member, I would like to view the future list that a book club plans to read.
48. As a reader, I would like to be able to scroll through available book clubs that are active for a book that I am currently reading.
49. As a reader, I would like to be able to form and name a book club.
50. As a reader, I would like to be able to comment and post within a club discussing a certain book that the group is currently reading.
51. As a reader, I would like to be able to report rude/hurtful comments and users.
52. As a reader, I would like to be able to mark a comment in a book club (including my own), as a spoiler if they have read past the other members of the book club.
53. As a moderator/team member, I would like to add books and their chapters to the catalogue.
54. As a moderator/team member, I would like to add summaries to books for display.
55. As a moderator/team member, I would like to add images of book covers for display.
56. As a moderator/team member, I would like for comments with sensitive content such as hate speech or slurs to be automatically flagged, and kept from being posted.
57. As a moderator/team member, I would like to be notified of reported comments.
58. As a moderator/team member, I would like to be able to delete reported comments.
59. As a moderator/team member, I would like to be able to allow a reported comment to be posted.
60. As a moderator/team member, I would like to be able to restrict users from commenting.
61. As a moderator/team member, I would like to be able to ban a user from the site.
62. As a moderator/team member, I would like to be able to mark a book club as inactive if no one has been active in the club for more than 6 months. (This can be reversed automatically when someone posts/comments in the club group)

Extra Features (If time allows):

63. As a book club member, I would like to discover publicly available book clubs from the books I'm currently reading.
64. As a reader, I would like to create summaries for each chapter of a given book and the most popular one (determined by likes) becomes the heading
65. As a reader, I would like to see books recommended to me based on my previous reading.
66. As a reader, I would like to log different reading sessions as I move through a book. In these sessions I would like to mark my beginning and end pages and mark what chapter I've most recently finished.

67. As a reader, I would like to mark where I am in a book via page numbers. (ex. Page number ____ of ____ pages, at ____%, with ___s being user input)
68. As a reader, I would like to be able to move my reviews/ratings from a different site onto this app with ease when transitioning to this app.
69. As a reader, I would like to access forums generalized to specific series or authors

---

Non-Functional Requirements

Performance

- The system shall respond to user requests within 500 ms for 95% of requests under normal load.
- The system shall respond to cached or repeated requests within 200 ms for 90% of cases.
- The system shall support up to 10,000 concurrent users without noticeable performance degradation.
- The frontend shall be designed to facilitate speedy and efficient content retrieval, minimizing client-side rendering times.
- API calls shall be optimized to reduce server load and data usage, including efficient pagination for large data sets.

Scalability

- The backend architecture shall be horizontally scalable to support usage ranging from a small number of users up to 10,000, with server upkeep costs scaling correspondingly.
- The database shall store only data functionally necessary for app functionality, in order to reduce server costs.
- The system shall support adding additional containers or services without requiring downtime.

Development / Architecture

- The frontend shall be developed with React Native to provide iOS and Android support.
- The backend shall include one or more ASP.NET Core Web APIs hosted by Azure Container Apps.
- The backend shall support containerized APIs, automatic scaling, and microservices.
- The system shall use Azure Cosmos DB as the primary document-based database.
- Additional Azure services shall be used as necessary to support core functionality (such asAzure App Insights for monitoring).

- The architecture shall be modular to allow independent updates of the frontend, backend, and database without breaking system functionality.

Security

- The system shall use Azure for backend hosting and storage of user profile and security information.
- The system shall support multi-factor authentication for user accounts. (if time allows)
- The development process shall follow DevSecOps practices, integrating security into each stage of development.
- Sensitive and necessary user data shall be encrypted at rest and in transit.
- Users shall only be able to view information on other users' profiles that has been explicitly marked as public. Any private information shall not be accessible or visible to other users.
- All authentication and authorization processes shall follow industry standards such as OAuth 2.0.

Usability

- The system shall remain launchable and operational at any time between now and the next two years.
- The system shall be distributable via Apple TestFlight, enabling installation through a shareable link.
- The system shall also be distributable via the Google Play Store.
- The user interface shall facilitate quick and intuitive interaction with core services.
- The app shall support accessibility features to accommodate users with disabilities, including screen reader compatibility and color contrast compliance.
- The app shall provide clear error messages and feedback to guide users when issues occur.

Reliability / Availability

- The system shall maintain 99% uptime under normal operating conditions.
- The system shall be able to recover from critical failures within 1 hour.
- Backups of the database shall be taken at least once every 24 hours and retained for a minimum of 30 days.
- The system shall continue functioning in a limited/offline mode if backend services are unavailable.

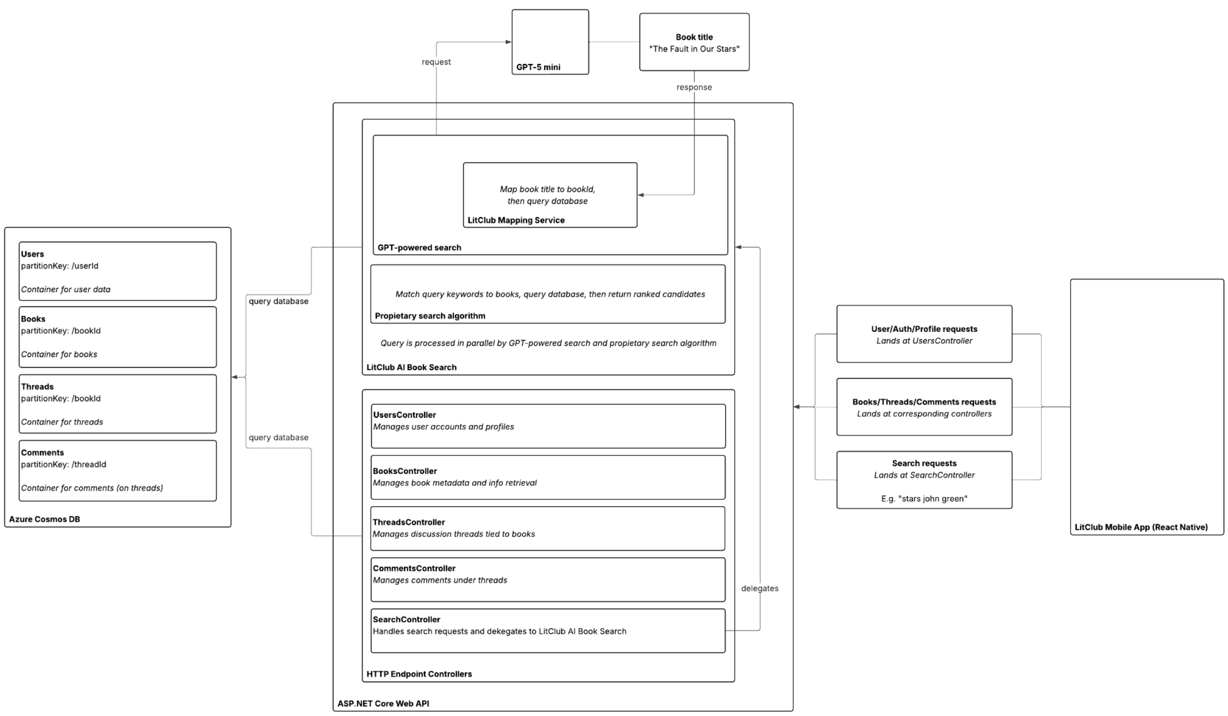
Hosting / Development Process

- Team members shall be able to access and modify a minimum of 90% of the codebase from a singular GitHub repository.
- The frontend and backend shall be manually testable independent from one another.
- Successful deployments shall be verifiable via integration tests that run automatically when merging into the develop branch.
- Team members shall be able to review code produced by other team members and approve or reject pull requests as appropriate.
- The team shall use Git/version control best practices, and the development tools and techniques applied shall support these practices.
- The CI/CD pipeline shall automatically deploy to a staging environment before production release

Design Outline

High Level Overview

The project derives from a client-server design, with a backend server and mobile client. The mobile client is made using React Native and the backend is hosted as an ASP.NET Core Web API. Information relating to the users is managed by UserController, and each respective function is handled by their respective controller. The AI-driven search feature for books allows us to map a gpt model to a book title using user input and maps the resulting book title to a bookid and a query database. All data that persists through each iteration and saves continuously is hosted via Azure Cosmos DB. All of this is then fed to the mobile client, hosted through React Native.



1. Data Models
    a. Formed using Azure Cosmos DB for containing and maintaining persistent data
    b. User
    c. Book
    d. Thread
    e. Comment
2. Controllers
    a. Hosted via ASP.NET Core Web API
    b. UserController
    c. BooksController
    d. ThreadsController
    e. SearchController

3. AI Search Layer
    a. GPT-Powered Search
    b. LitClub Mapping Service
4. Mobile App
    a. Hosted through ReactNative, client side

<u>Design Issues</u>

Functional Issues

1. What information is required for creating a user account?
    a. Name, Email/Phone Number, Password
    b. Name, Email, Phone Number, Username, Password
    c. Name, Email/Phone Number, Username, Password

    Option: C

    Justification: We chose this option because this is most consistent with the existing sign up information that other apps and services use to make accounts. Additionally, we believe a lot of the requirements make natural sense, such as requiring a username and password to log in, a name for the user's profile, and an email or phone number for recovery information.

2. What happens if a chat or forum has offensive language or hurtful comments?
    a. Users can report a comment or review that they believe is harmful and will be flagged and reviewed by moderators.
    b. Comments containing pre-defined hurtful or offensive language will be restricted from being posted.
    c. Flag users who consistently post hurtful or damaging messages/comments/reviews on the moderator side.

    Option: All of the above

    Justification: We believe that harassment, bullying, or harmful language is extremely harmful to users of the app and directly conflicts with our team's moral obligations to design software that does not harm society. As a result, we ensure that all of the above actions are taken to mitigate any and all forms of offensive language and harassment/bullying.

3. Do users need to add a book to their shelf to read comments or discussions for said book?
    a. Yes; Users must have added a book to read and post comments.
    b. Depends; Readers must have added a book to their shelf to post comments, but not to read them.

   c. No; Readers can both read and post comments for books that are not on their shelf.

Option: A

Justification: Part of our philosophy and intended use of LitClub is that users will not receive spoilers and will feel as though they are reading and reacting to a book alongside the community. Allowing users to see and post comments on books that they have not read and may not have an intention to read breaks this illusion. If a user would like to break the illusion for themselves or has genuine interest in reading the book, they will add the book to their bookshelf anyways, so this choice protects the users who wish to stay immersed in LitClub from spoilers.

4. What happens if a user's comments/updates on a book don't save/they are not connected to WiFi?
   a. User comments and book updates can only be saved when connected to WiFi.
   b. User comments can only be saved when connected to WiFi, but updates to their bookshelf can be done without internet connection.
   c. Both comments and bookshelf updates can be saved without WiFi connection.

Option: A

Justification: Because of the social focus on our app, users would not have offline abilities within LitClub. Other users would need to be able to see the changes that a user makes to their own personal library as a part of their profile, therefore the information must be stored in the server and updated while a user is connected to the internet.

Non-Functional Issues

1. What framework is best suited for our app's needs?
   a. React Native
   b. Flutter
   c. Kotlin
   d. Swift

Choice: React Native

Justification: We decided to use React Native because we are looking to make our app cross platform, and our group members already have some experience working with frameworks similar to it. Kotlin and Swift are both powerful languages for Android and IOS development respectively, but we want a cross platform approach, so we had to rule them out. Flutter is a popular option for cross platform development, but our team does not have any experience using it.

2. What cloud computing service is most ideal for implementing our backend services?
   a. Google Firebase

   b. Microsoft Azure
   c. AWS

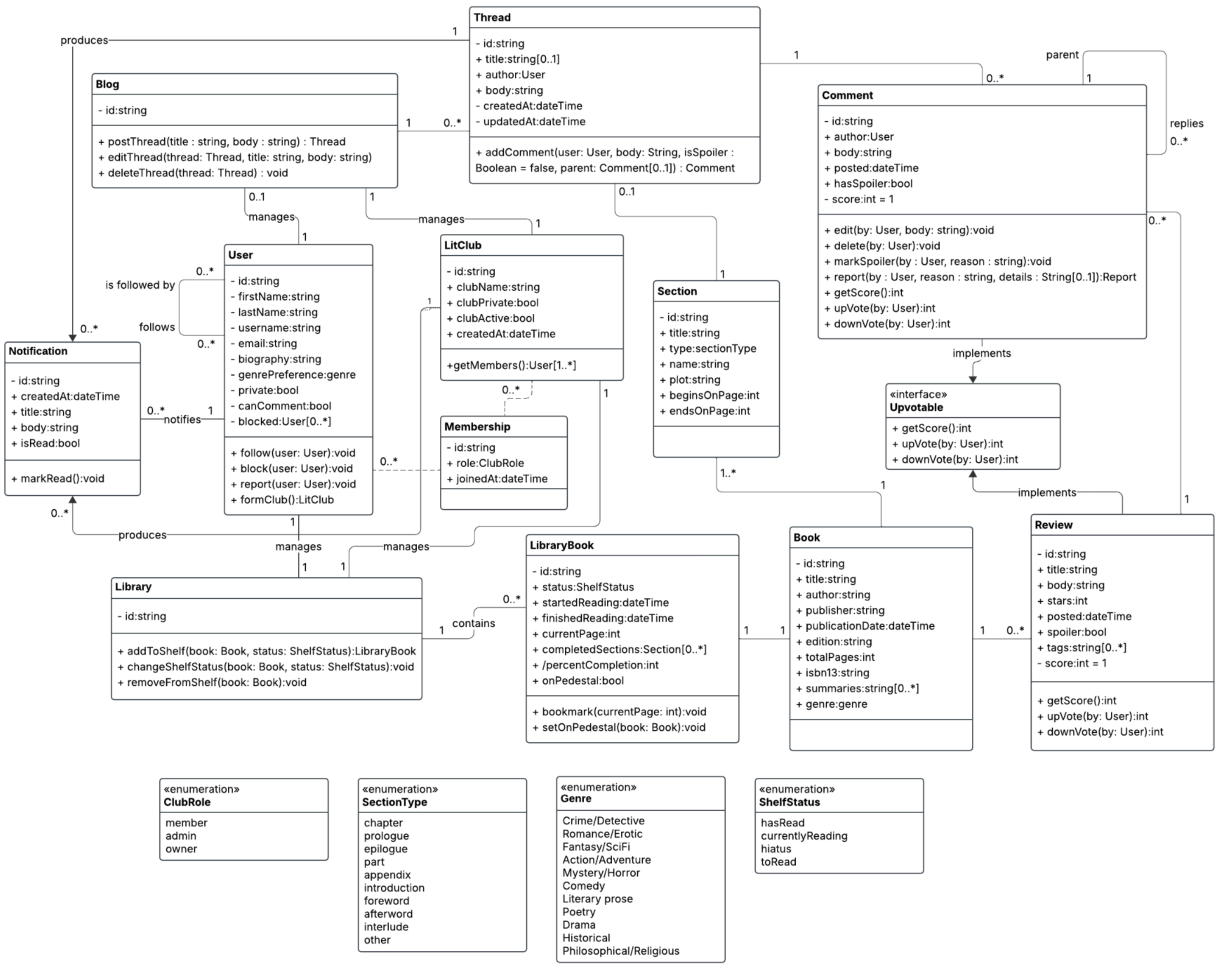
Choice: Microsoft Azure
Justification: We decided to use Azure because our primary backend developer in our group has experience with this platform and credits to spare for developing with it. Our group has agreed to continue paying for that cost once those credits expire, as it is inexpensive. Azure offers automatic scaling depending on usage amount, and we can use Microsoft Entra ID (AAD) for user security purposes.

3. How will data be stored?
   a. Azure Cosmos DB
   b. MongoDB
   c. Oracle Database
   d. Google Firestore

Choice: Azure Cosmos DB
Justification: Cosmos DB is already heavily integrated into Azure infrastructure, so there is already incentive to choose it. In addition, Cosmos DB is elastic in its scaling and offers very low latency. The Database is globally distributed, allowing users to interact with it with low latency from anywhere.

Design Details

Class Level Design



Description of Class and their Interactions

- **User**
  - A user manages their personal library of books
  - A user may have a personal blog
  - A user can follow and be followed by other users
  - A user can receive notifications produced by the system
  - A user can join a LitClub and be a member, admin, or owner
- **Library**
  - A library is managed by a user (personal) or LitClub (club library)
  - A library contains many library books
- **LibraryBook**
  - The library book is associated with a book in the database

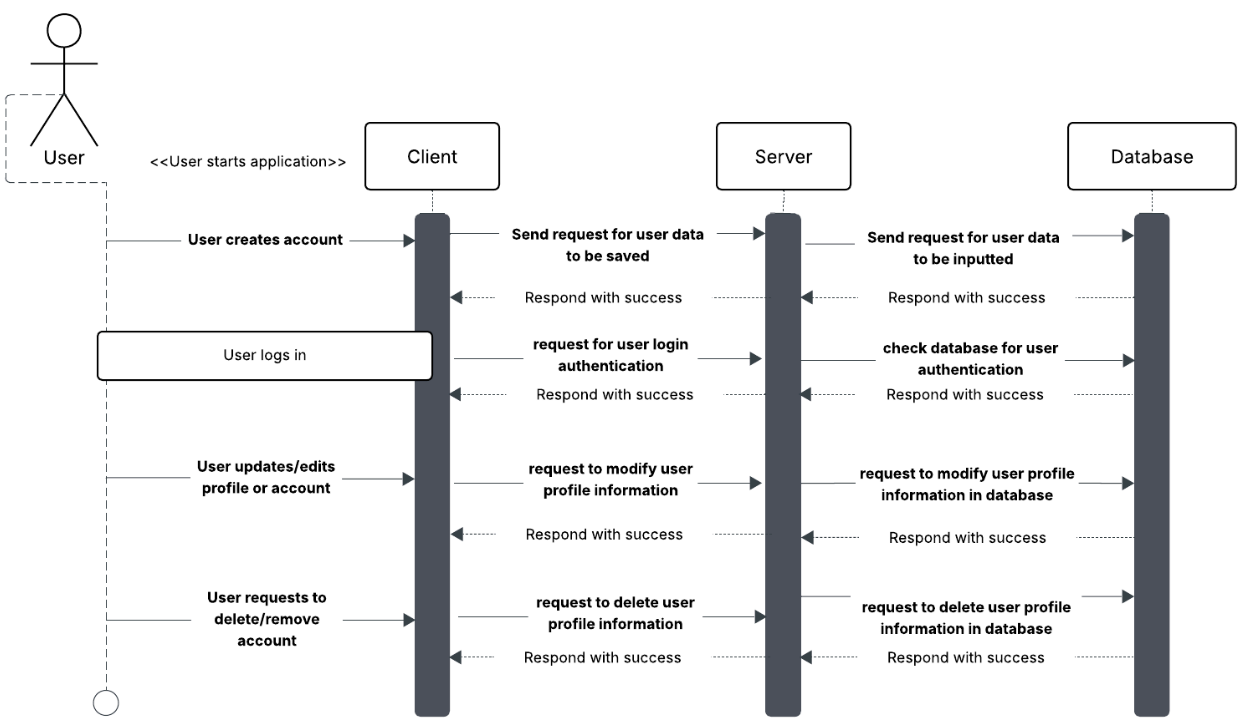
  - A library book stores the user's reading state for a specific book
- **Book**
  - The book contains book metadata and has many reviews
  - A book contains many sections
- **Review**
  - A review contains a title, body, stars, etc., and is authored by a user
  - A comment can be upvoted or downvoted by users
  - A review can have many comments
- **Section**
  - A section has a title, beginning and ending pages, and a plot summary
  - A section is described by SectionType
  - A section can have zero or one discussion threads
- **Thread**
  - A thread is associated with either a blog or a section
  - A thread has a title and an author (if associated with a blog), and a text body
  - A thread can have many comments
  - Activity in a thread can trigger notifications
- **Comment**
  - A comment can be made on a thread or a book review
  - A comment can be made on other comments
  - A comment can be upvoted or downvoted by users
  - The contents of a comment can be edited or deleted
  - A comment can be marked as a spoiler by a user
  - A comment can be reported by a user for various reasons
- **LitClub**
  - A LitClub has one or more members
  - A LitClub has a name
  - A LitClub can be made public or private
  - A LitClub manages a club library
  - A LitClub manages a club blog
  - A LitClub can be marked active or inactive
  - A LitClub stores its creation timestamp
- **Blog**
  - A blog contains zero to many threads
  - A blog is managed by a user or a LitClub
  - Threads can be created, edited, and deleted on a blog
- **Notification**
  - Notifications are triggered by events which occur in threads and LitClubs
  - A notification is delivered to a user recipient
  - A notification has a title and text body

- A notification has a timestamp for its creation
- A notification can be marked as read

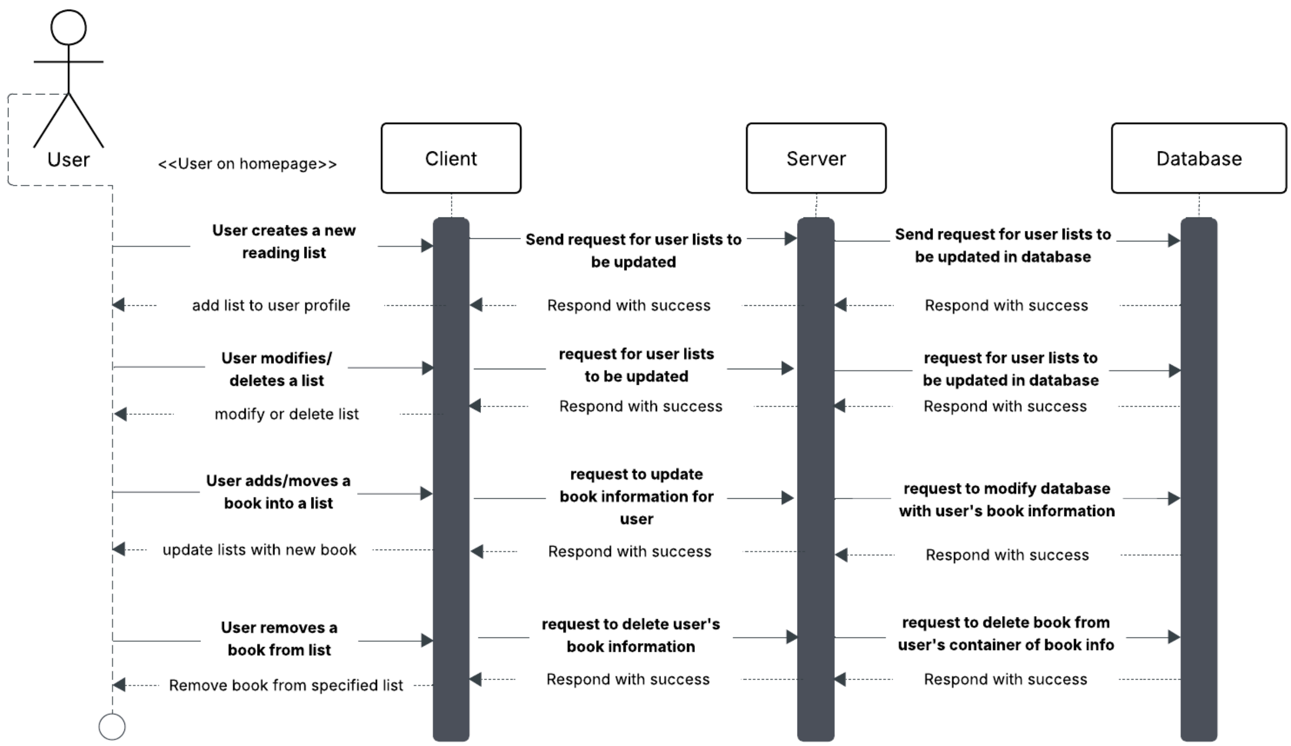
Sequence Diagrams

The sequence diagrams show the sequence of events that actions will be carried out in once the user launches the app. It shows the typical flow of interaction between the mobile client, backend server, and database. As the user logs in to the app, the client sends a request to the backend server, hosted by ASP.NET Core Web API. From there, the server connects to the database, which has separate containers of information for each aspect of information, from user information, book information, and club and group information. The user, when they make an account, sends a request to the server which then connects to the database and adds a user profile. From there, whenever edits need to occur, the server sends a request to the database to perform a change. Likewise, for all sequences of events, when the user attempts to add, modify, or delete a feature of the app, the client processes that action and sends a request to the server, which then connects to the database and takes the necessary action to complete the step.

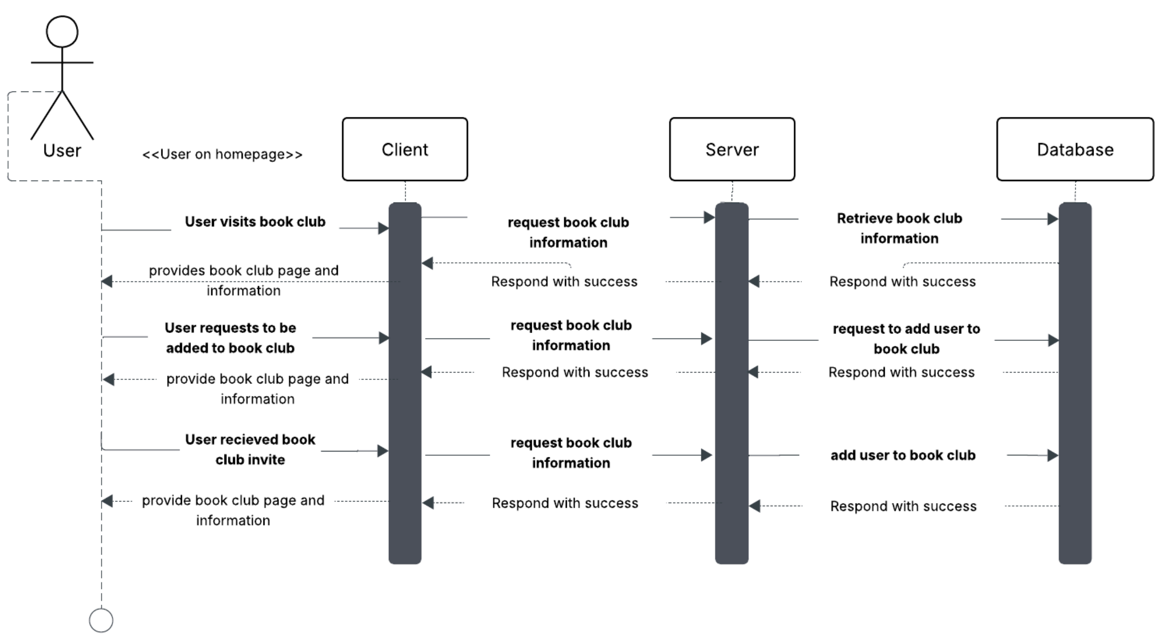
User Account Set Up, Login, and Profile Information



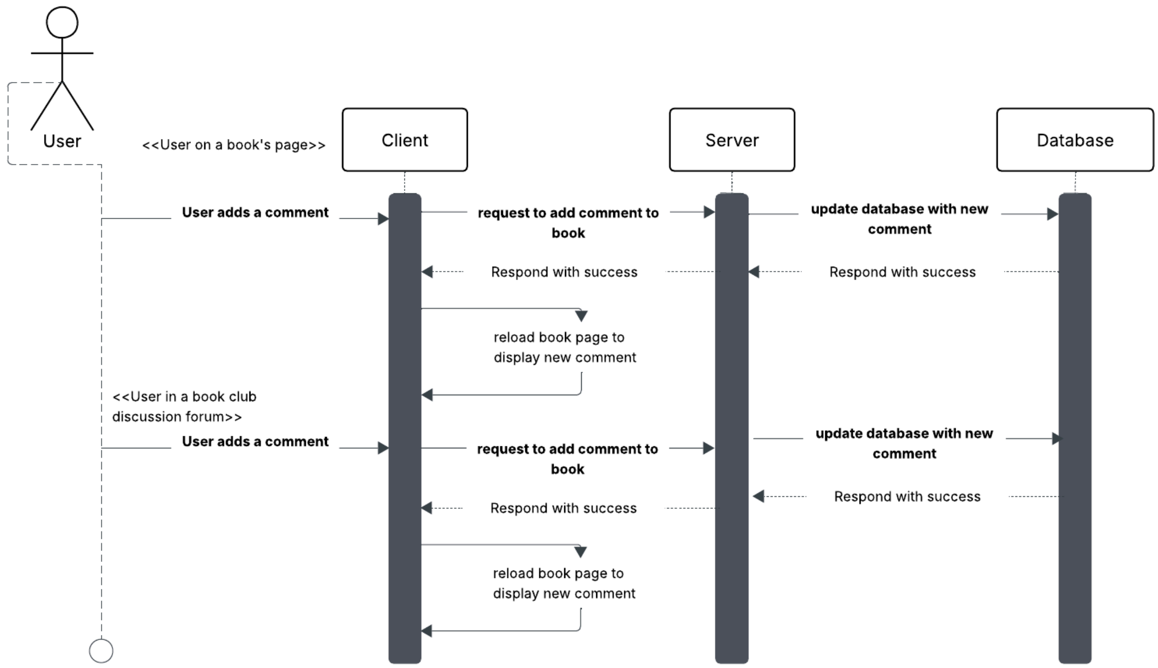
User Request to Add/Modify/Delete Library Books and Reading Lists



User Request to View or Join a Book Club



User Writes a Comment on a Book Forum or Book Club Discussion Thread



Navigation Flow Map
In order to best suit LitClub's functions to user needs, our team decided to focus on mobile development. While we believe a web version may have certain perks, most users opt for logging reading habits on their phone. Additionally, most social media is often consumed in a mobile format, further solidifying our choice to focus on mobile development. The app will be user-friendly, allowing for easy use and a functional design.

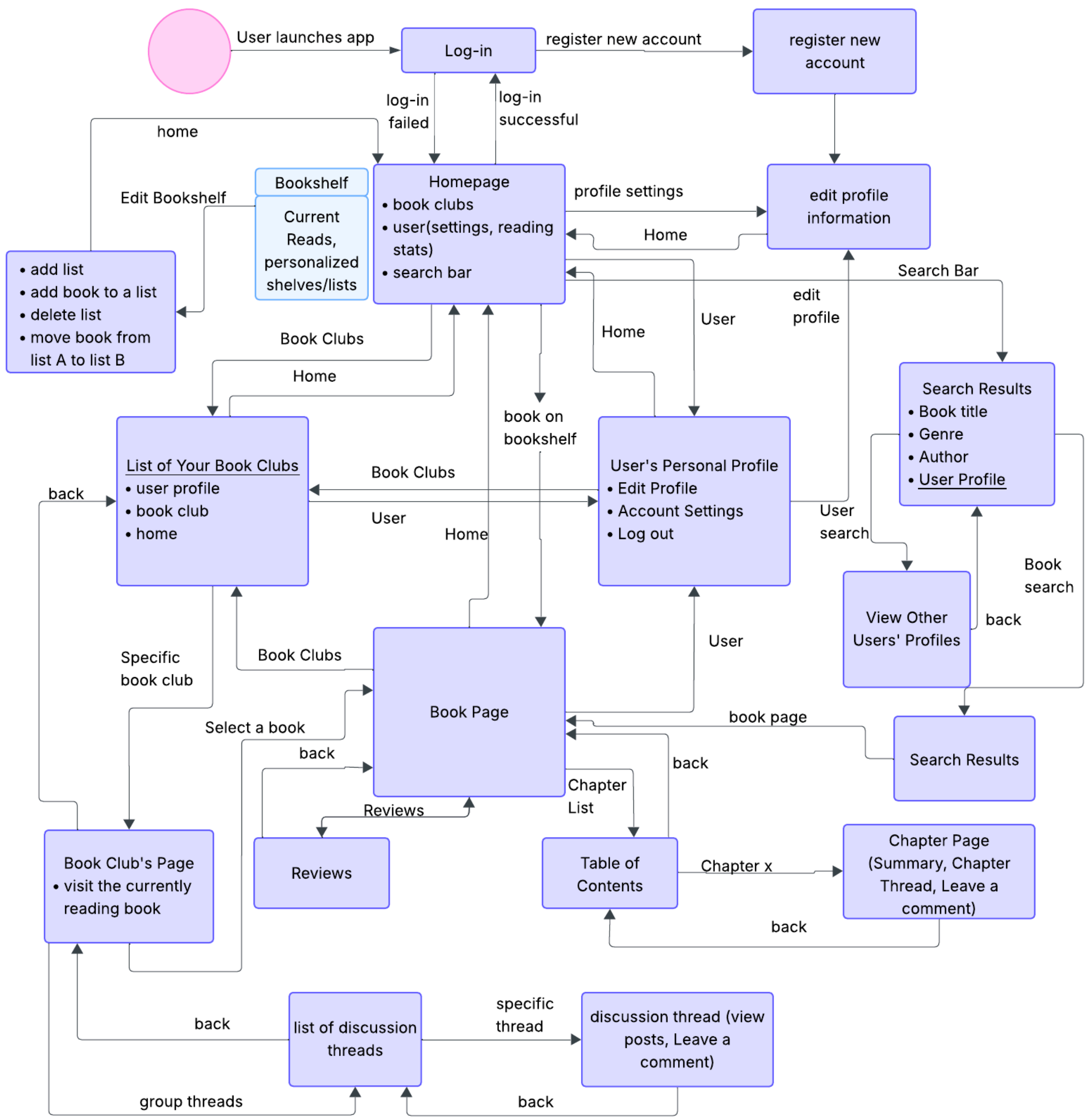
When the user first launches the app, they are prompted to log in or sign up. If they have previously already signed in and chose to remain signed in, they will skip this step. The homepage consists of the users bookshelf, which has all their lists– including their current reads.

At the bottom of the screen, there are three buttons to go to three different pages: the home page, as listed above, the book clubs page, and a user account page. The book club pages take the user to their list of their current book clubs along with any notifications and updates from their clubs. From here, users can also join new clubs and leave old ones.

The user account page allows the user to edit their profiles, see their reading stats, and edit any account information.

The user can also use the search bar to search for books by author, title or genre. They can also search for other users or bookclubs. This search bar is accessible from most pages to allow ease for the user.

Additionally, a user can add reviews or comments to a chapter thread through visiting a book's page. They can access book pages from a search, from their bookshelf, or from a book club that they are in. Additionally, users can add comments to the discussion forums exclusive to their book clubs.



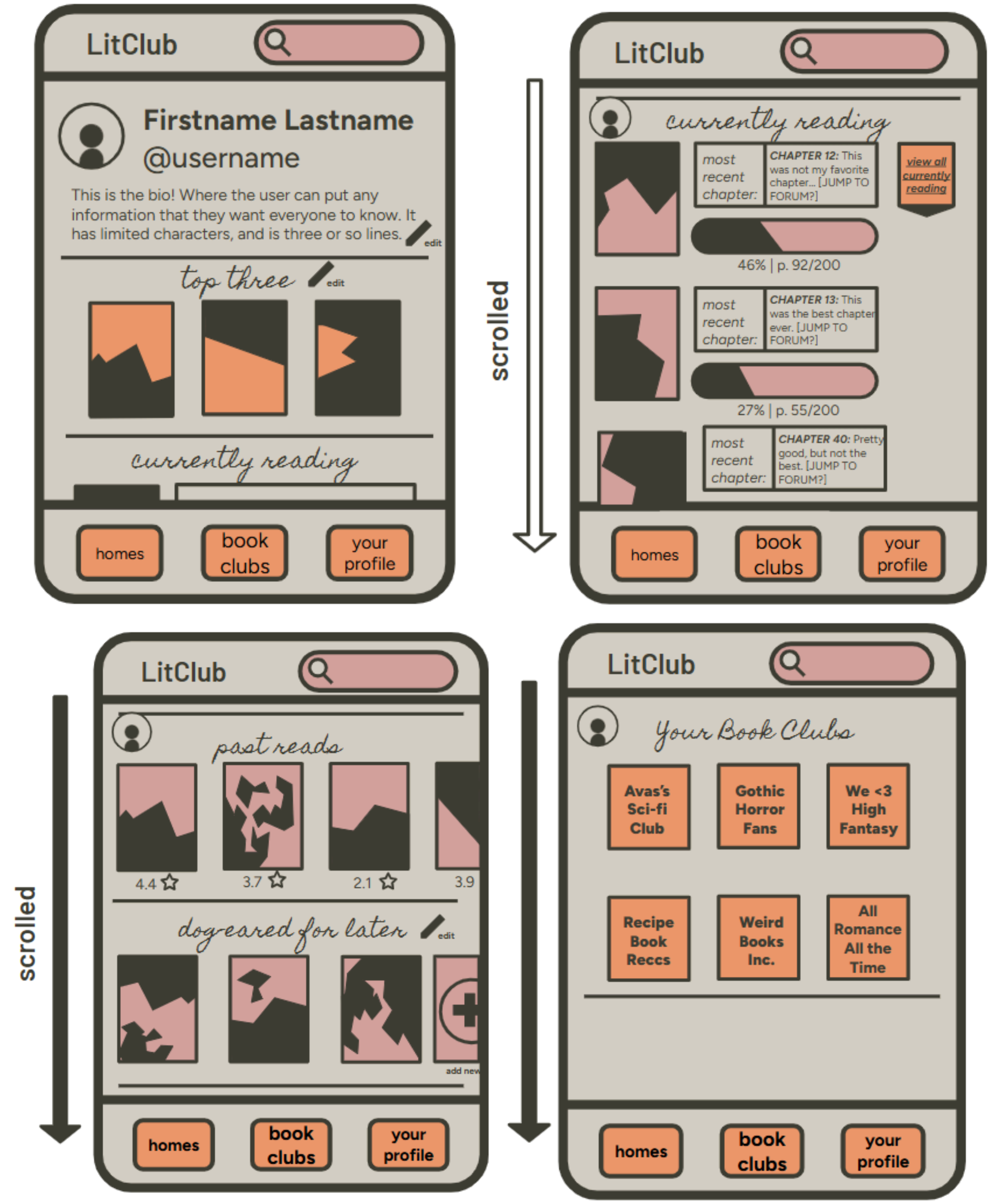
UI MockUp:

Home Screen

Log Session Screen

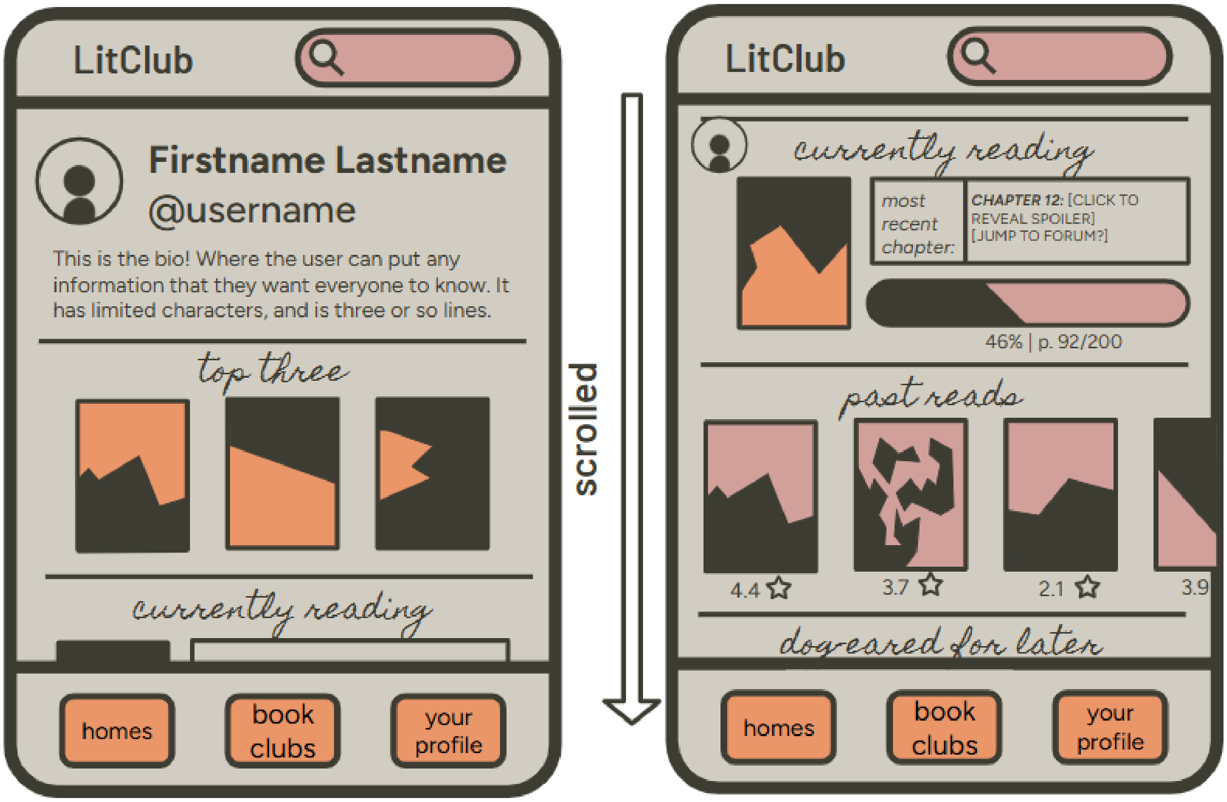
The home screen is the landing screen that will show up immediately when the user opens the app. It has information on the books they're currently reading, as well as their to-be-read list. From here, they can navigate to their profile, to view their book clubs, or to the search screen. They can also hit 'Log Reading Session' to navigate to that screen, in which they can log a reading session of their most recently read book.

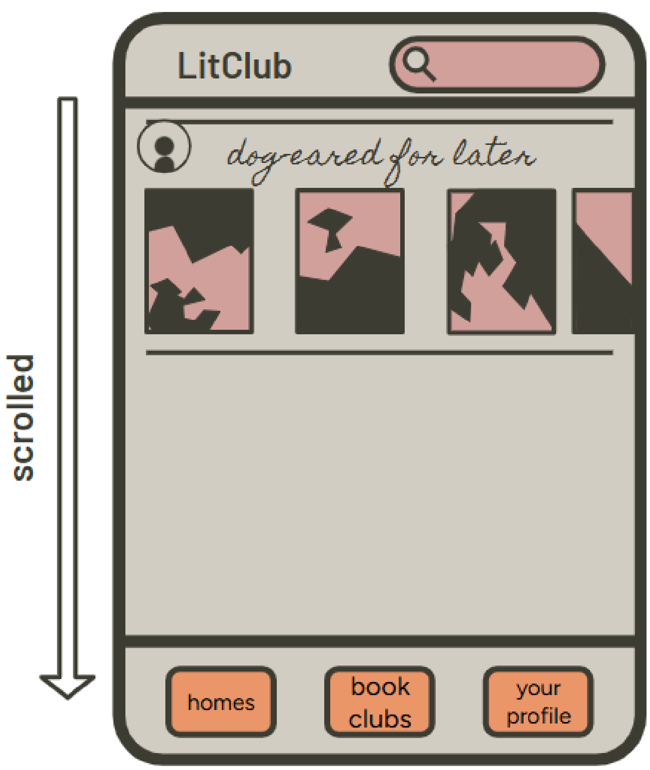
View Your Profile



This is the page for a user's own profile. It allows easy access to view and edit all of their information. Clicking on a book will take the user to that book's page, where they can view the chapter forum and reviews for the novel. The currently reading page is expandible, and when collapsed only shows the most recent book logged.

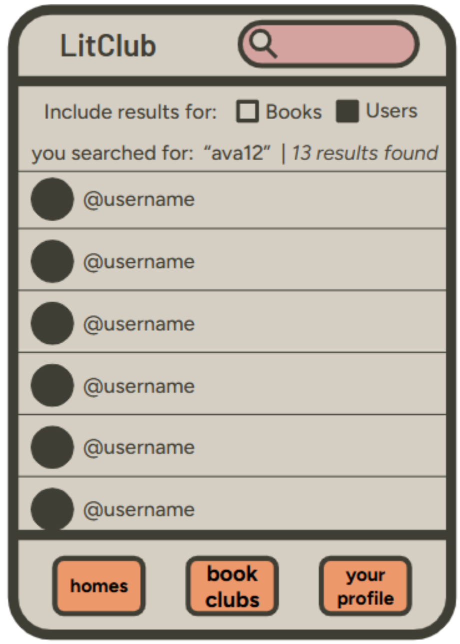
Viewing Another's Profile





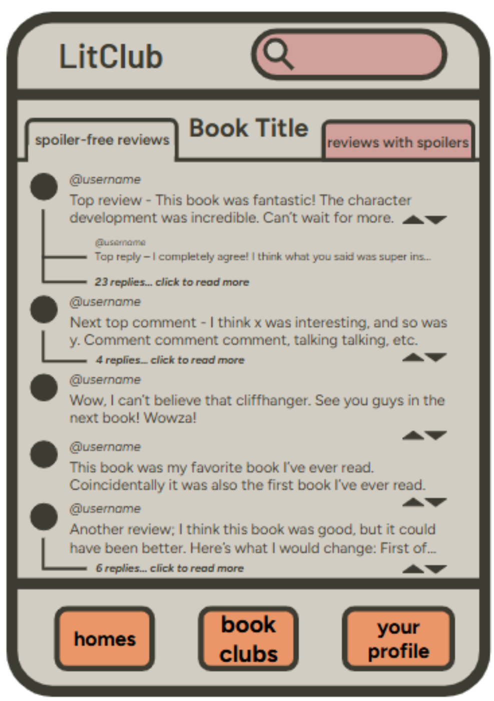
A user will be able to access another's profile by clicking on their icon or username. This is a similar screen to viewing your own profile, but they will not be able to edit anything, just click on items.

Search Results



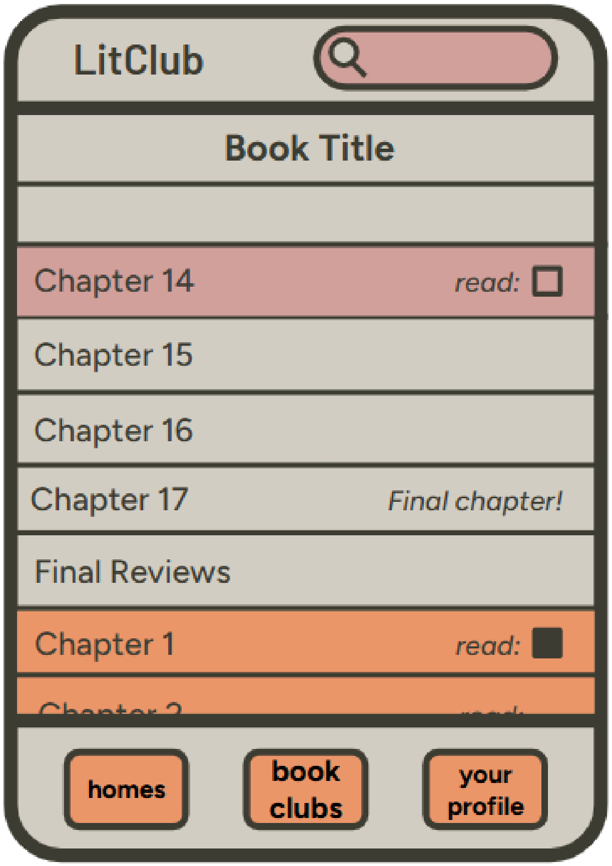
Users will have the option to search for either books or users, and will be able to see different results upon searching.

Book Page **Reviews Page**

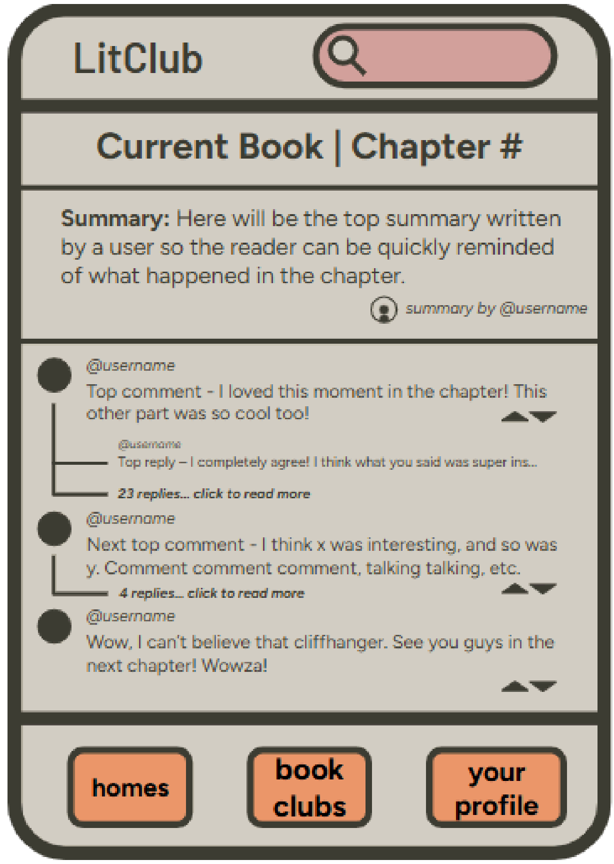


The book page will be the page users are taken to when they click on a book. From here, they can view the reviews (either with or without spoilers) or navigate to the table of contents to access the forums.

Table of Contents/Chapter List

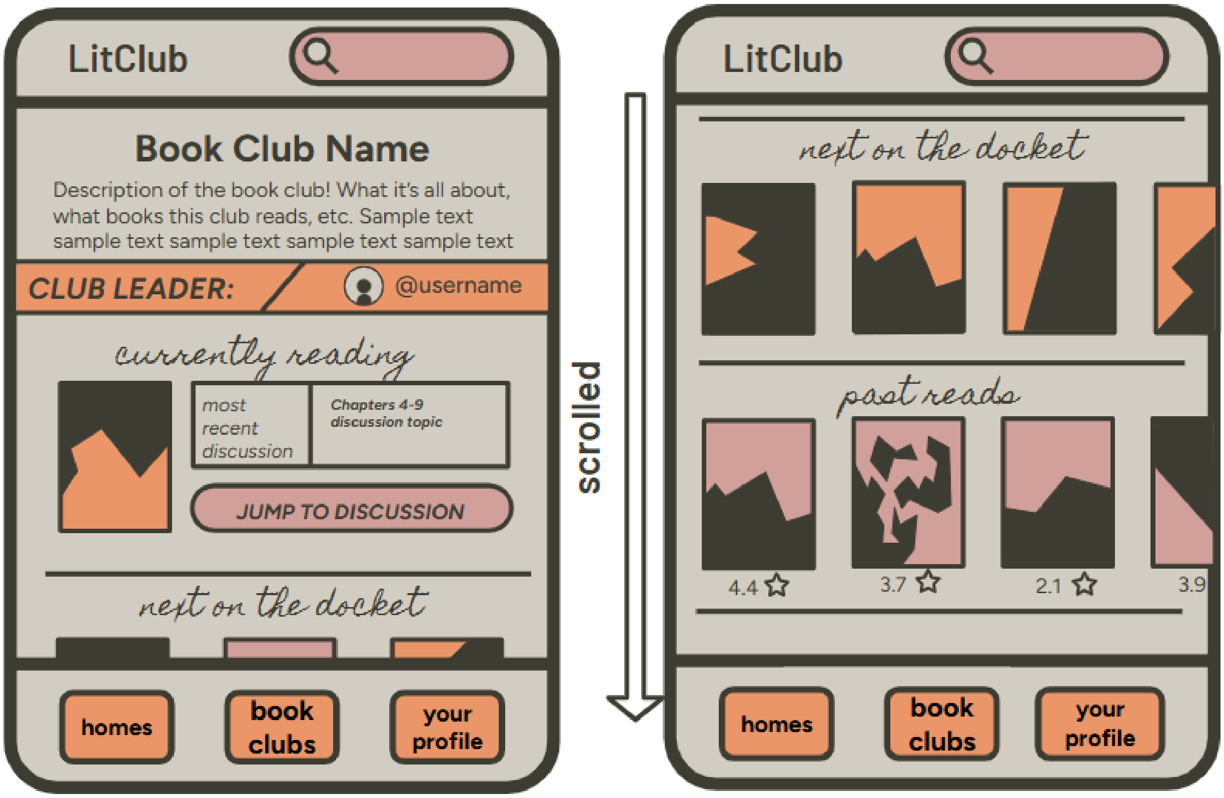


Individual Chapter Thread



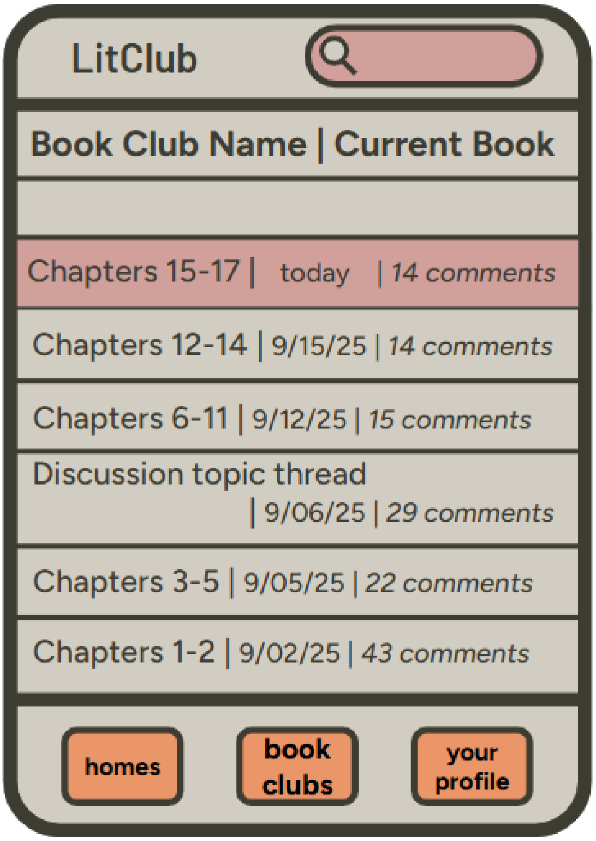
This is what the app takes the user to when "table of contents" is clicked on. The top chapter is the next chapter for the user to read, tracked via their reading session logs. Chapters in orange are ones the user has already read. When a chapter is clicked on, the user can view the comments on the individual chapter. In the chapter thread, users can expand or collapse comment chains, and upvote or downvote individual comments.

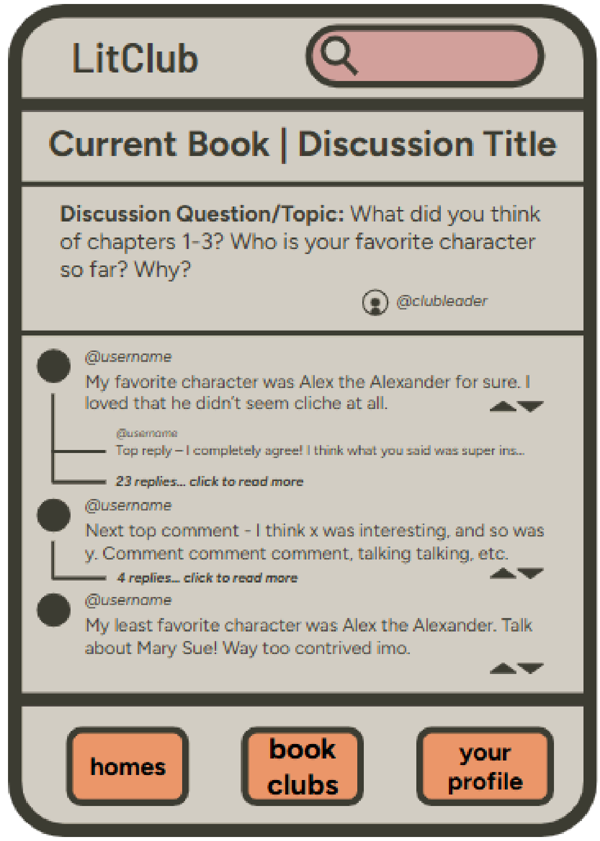
Book Club Page **(scrolled down)**



Here is where it takes you when the user clicks on one of their book clubs. They can view the club's current book, as well as see the current topic of discussion. They can also see books that will be read soon and previous reads of the club.

List of Book Club Discussion Threads **Individual Discussion Thread**





This is the main function of the book club: setting a pace for the readers and creating discussion threads based on that pace. The leader can mark specific chunks of the book for the members to read, and then they discuss that chunk specifically. The functionality is very similar to the individual chapter thread functionality.